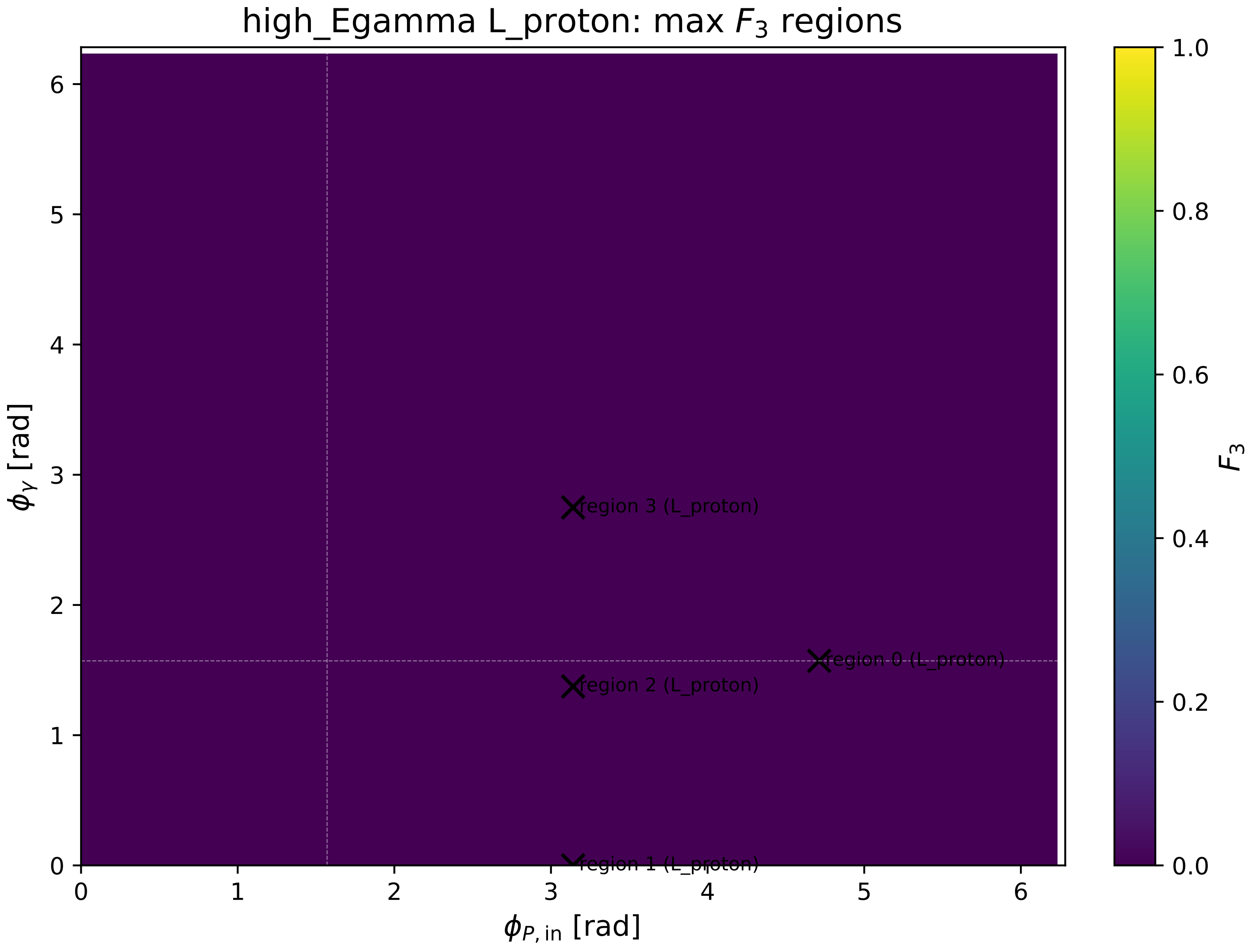
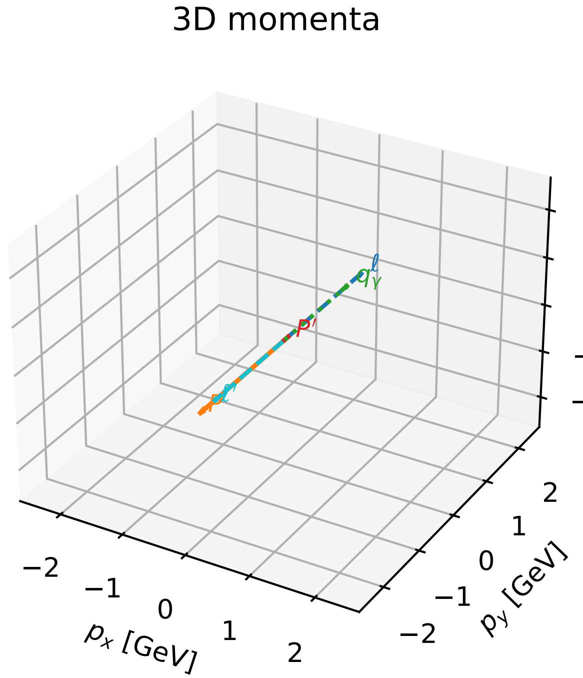

high_Egamma L_proton: max F_3 regions
region 3 (L_proton)
region 0 (L_proton)
region 2 (L_proton)
region 1 (L_proton)
$\phi_{P,\mathrm{in}}$ [rad]
ϕ_γ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L_proton characteristic max F_3 configuration

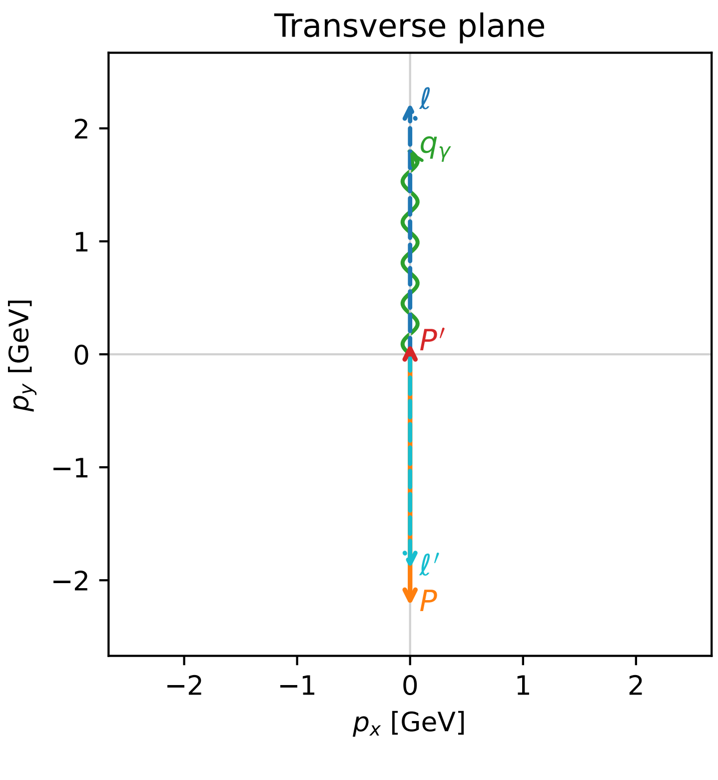


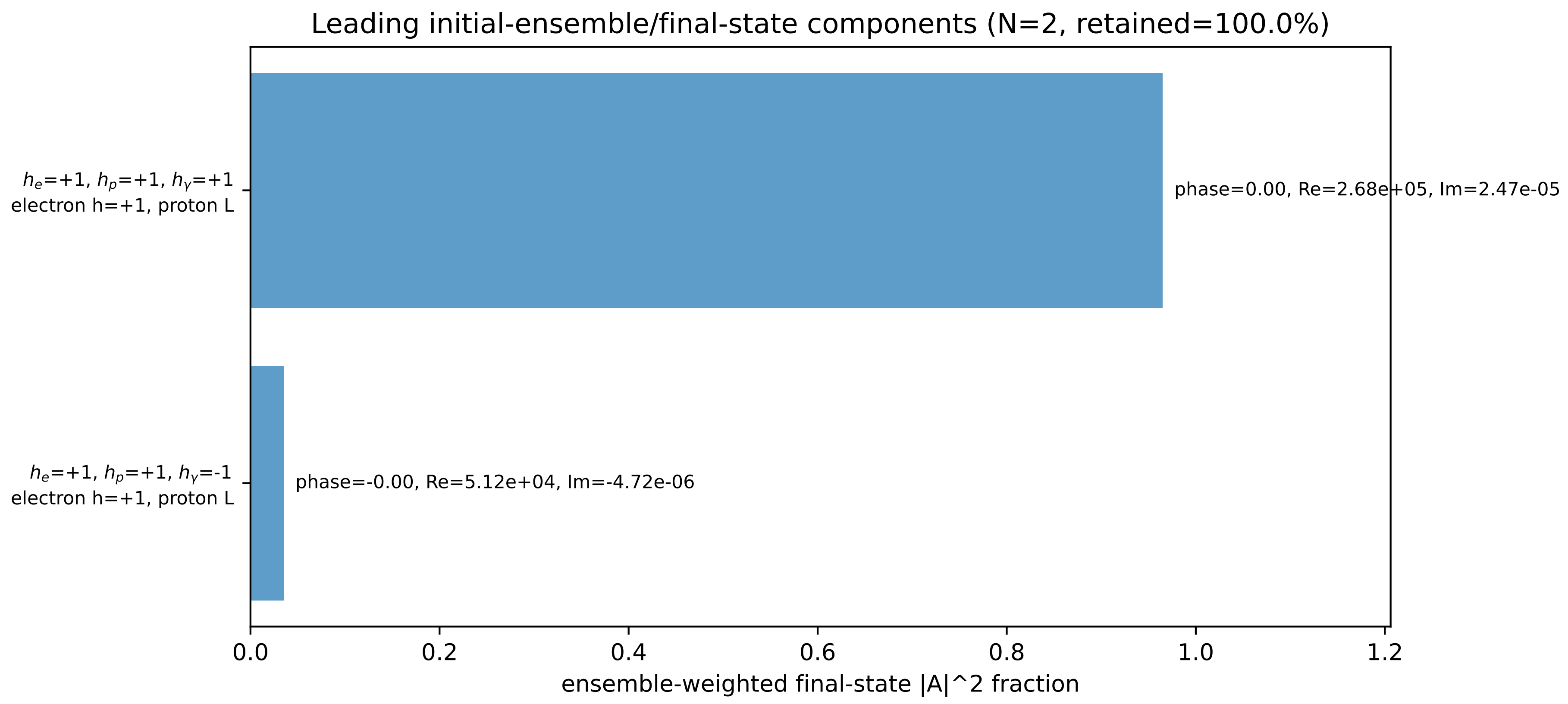

```
f3_high_egamma_l_proton_region_0  F_3=1.86023e-14
region: high_Egamma_L_proton_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

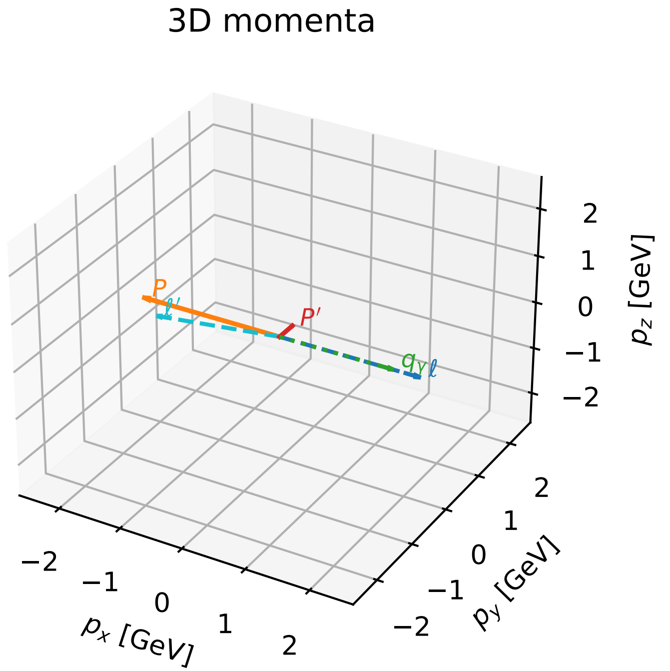
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.0956529
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=1.5708
Q^2=16.8669, x_B=0.846865, t=-3.21819, W^2=3.92981, y=0.965151
$\theta(\ell', \gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 1.8957 p_x= -0.0000 p_y= -1.8957 p_z= 0.0000
P' E= 0.9429 p_x= 0.0000 p_y= 0.0957 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.0000 p_y= 1.8000 p_z= 0.0000





L_proton characteristic max F_3 configuration

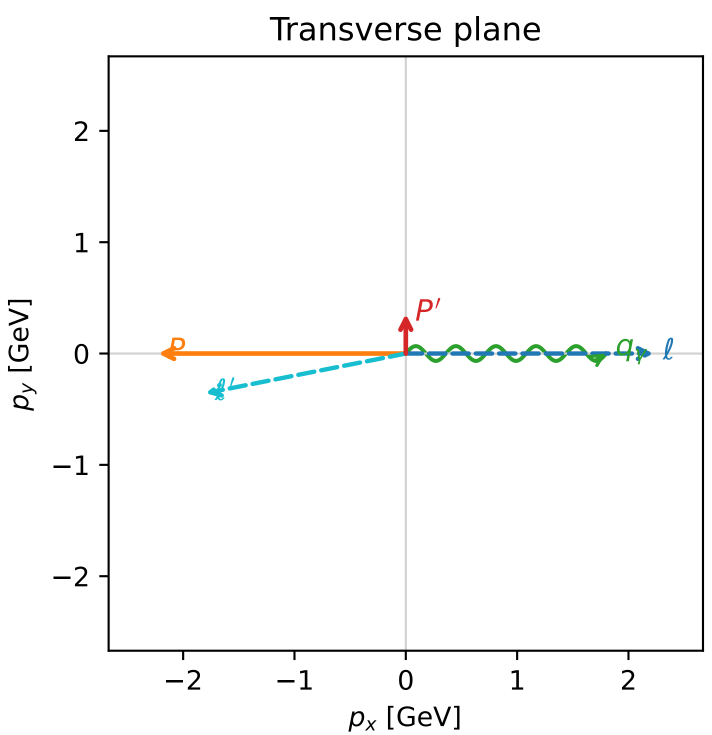


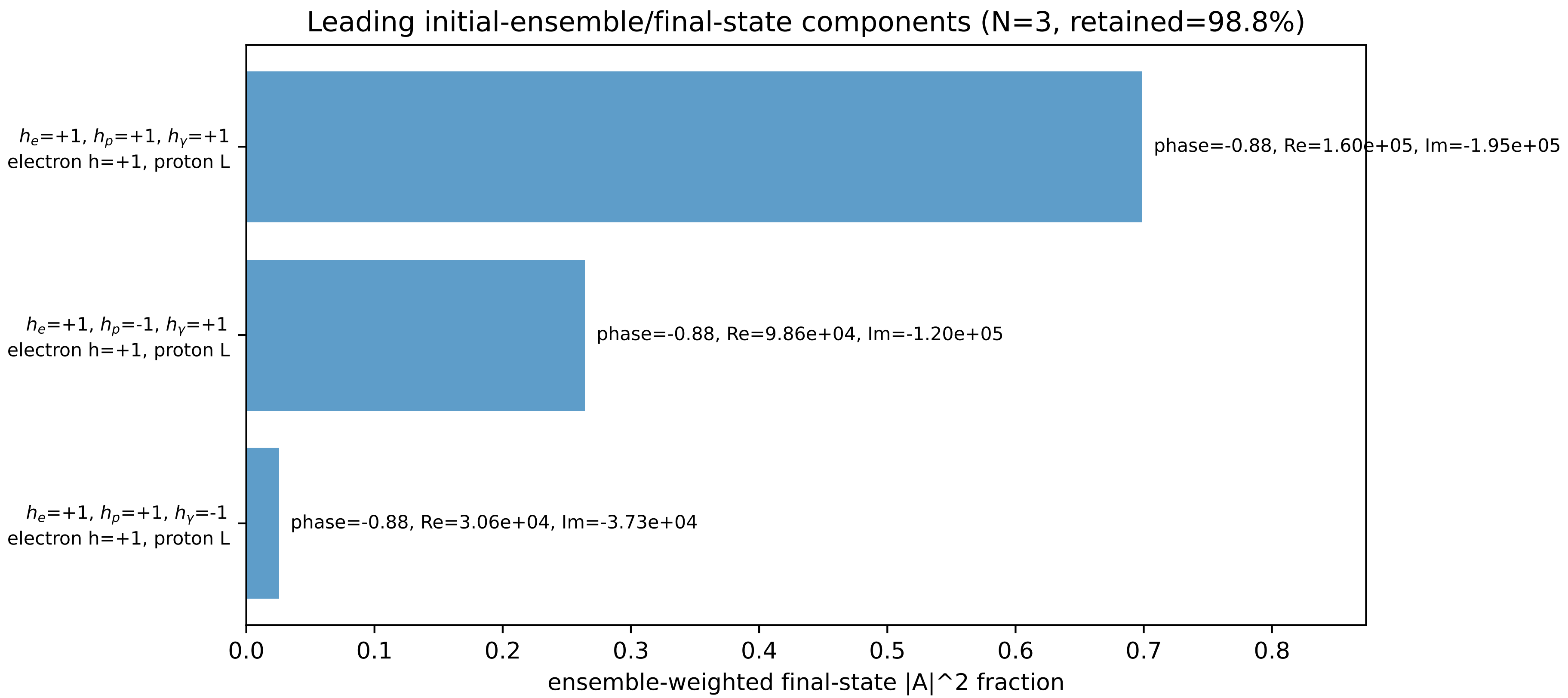

```
f3_high_egamma_l_proton_region_1  F_3=0
region: high_Egamma_L_proton_region_1
outgoing-state purity: 0.995661
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.356666
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=0
Q^2=16.1715, x_B=0.817396, t=-3.08551, W^2=4.49252, y=0.958721
$\theta(\ell', \gamma)$=168.792 deg

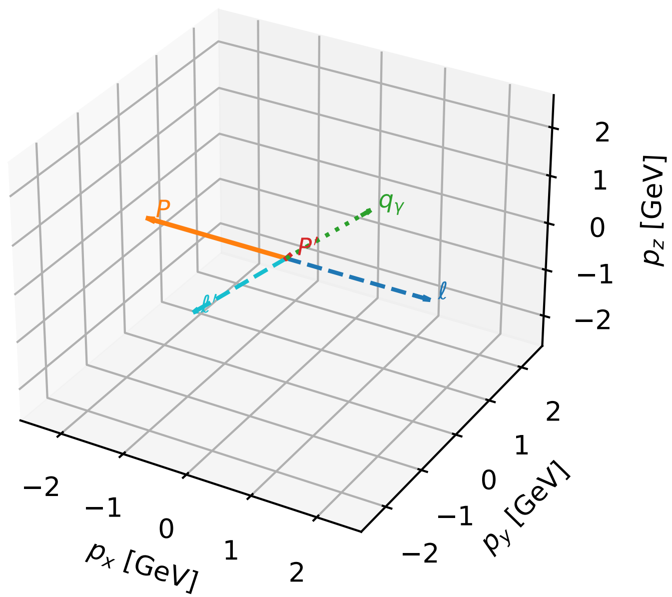
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8350 p_x= -1.8000 p_y= -0.3567 p_z= 0.0000
P' E= 1.0035 p_x= 0.0000 p_y= 0.3567 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.8000 p_y= 0.0000 p_z= 0.0000





L_proton characteristic max F_3 configuration

3D momenta

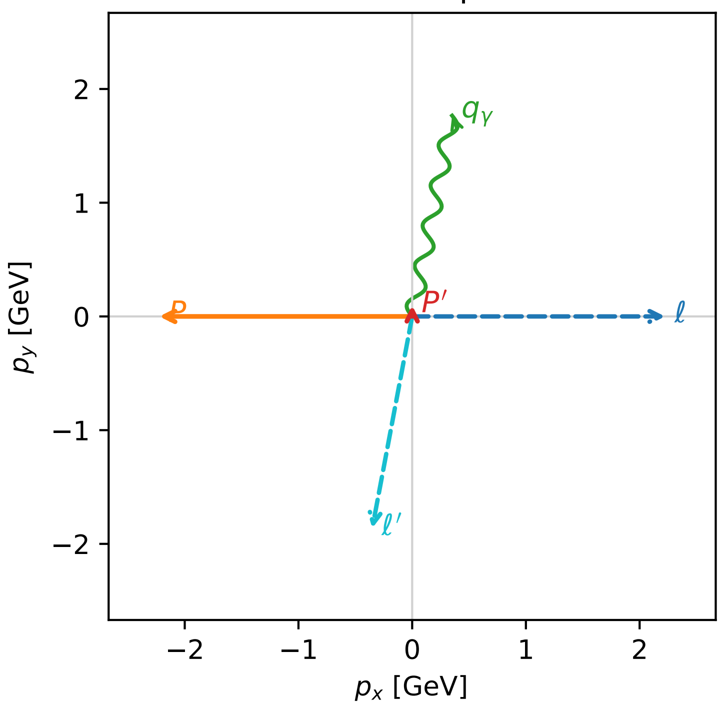


f3_high_egamma_l_proton_region_2 F_3=0
region: high_Egamma_L_proton_region_2
outgoing-state purity: 0.809527
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

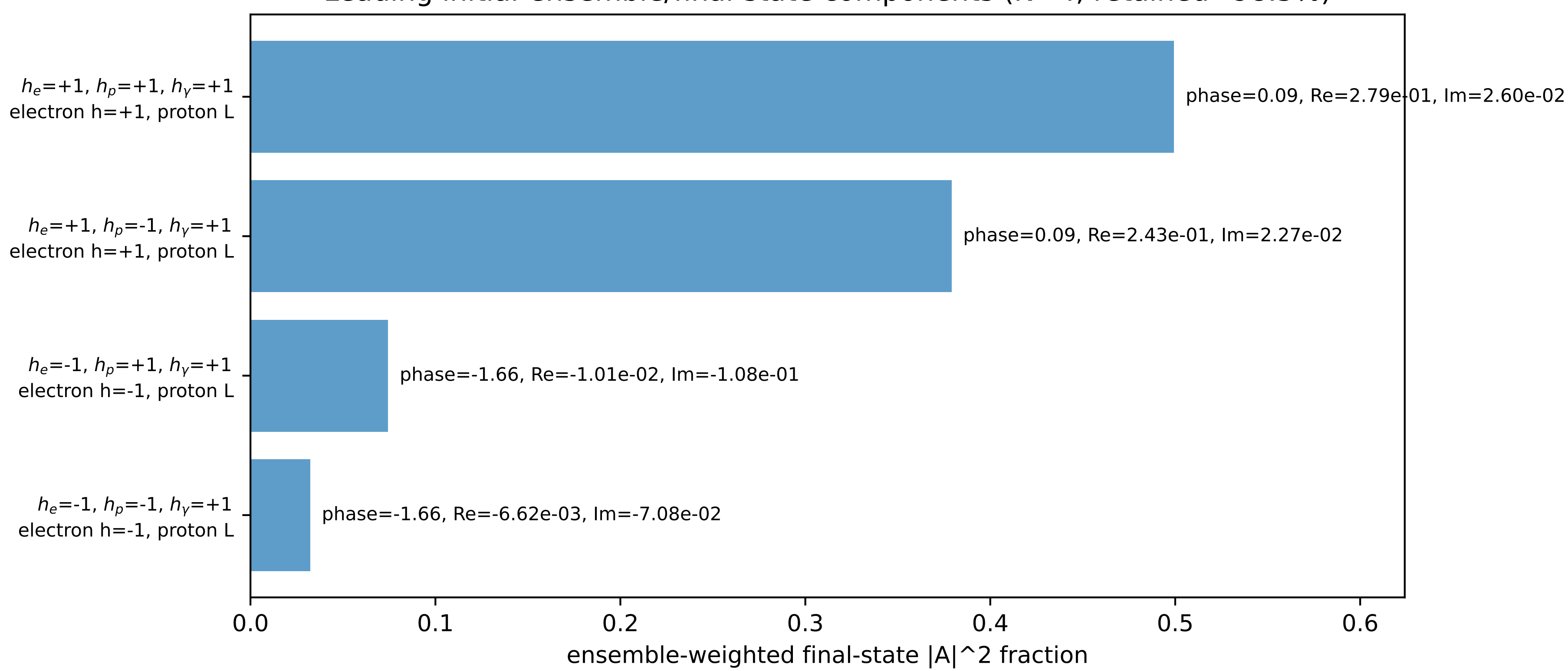
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.0972622
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=1.37445
Q^2=9.99498, x_B=0.766106, t=-2.79344, W^2=3.93133, y=0.632219
$\theta(\ell', \gamma)$=179.426 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8955 p_x= -0.3512 p_y= -1.8627 p_z= 0.0000
P' E= 0.9430 p_x= 0.0000 p_y= 0.0973 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.3512 p_y= 1.7654 p_z= 0.0000

Transverse plane

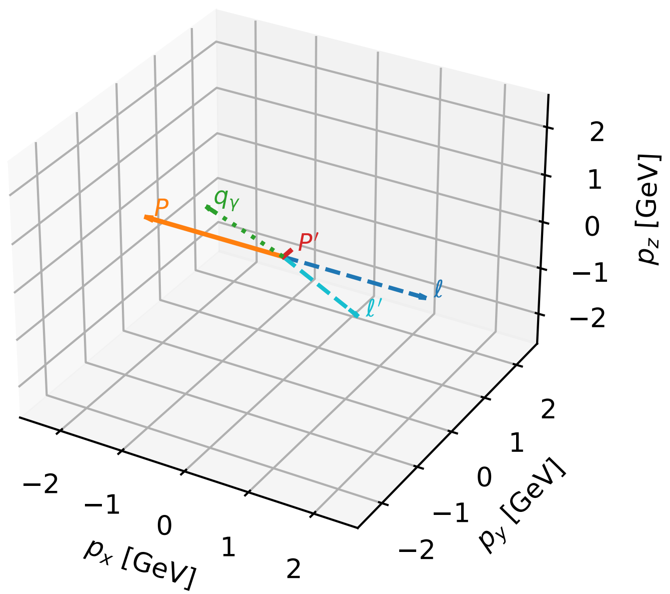


Leading initial-ensemble/final-state components (N=4, retained=98.5%)



L_proton characteristic max F_3 configuration

3D momenta

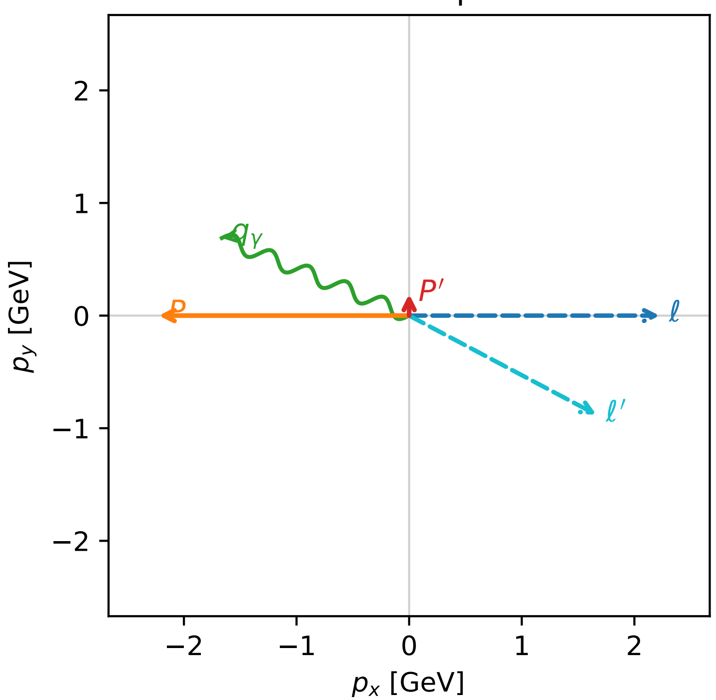


f3_high_egamma_l_proton_region_3 F_3=0
region: high_Egamma_L_proton_region_3
outgoing-state purity: 0.525465
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.190824
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=2.74889
Q^2=0.971273, x_B=0.233797, t=-2.86193, W^2=4.06292, y=0.201316
$\theta(\ell', \gamma)$=174.623 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8813 p_x= 1.6630 p_y= -0.8797 p_z= 0.0000
P' E= 0.9572 p_x= 0.0000 p_y= 0.1908 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.6630 p_y= 0.6888 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.9%)